Database Management Systems

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Session Objective

- DBMS Introduction
- DBMS - An example
- Characteristics of Database Approach
- Database Users

Session Outcome

At the end of this session, participants will be able to:

- Understand the basis of DBMS
- Understand the Characteristics of database
- Understand simplified database environment
- Database Users

Table of Contents

1. Introduction- Database
2. Database Environment
3. Characteristics of database Approach
4. Database-users
5. Summary
6. Reference

Introduction

Database and Database technology had a major impact on the growing use of computers.

- **Data:** Known facts that can be recorded and have an implicit meaning
- **Database:** A very large, integrated collection of interrelated data.
- Eg: Contact list in Mobile phones

Properties of database

A database has following implicit properties:

- A database(db) represents some aspects of the real world called mini world and changes in the mini world are reflected in the db.
- It is a **logically coherent collection of data** with some inherent meaning but not a random collection.
- It is **designed, built, and populated** with data for a specific purpose.
- It has an intended group of users and some preconceived applications in which users are interested.

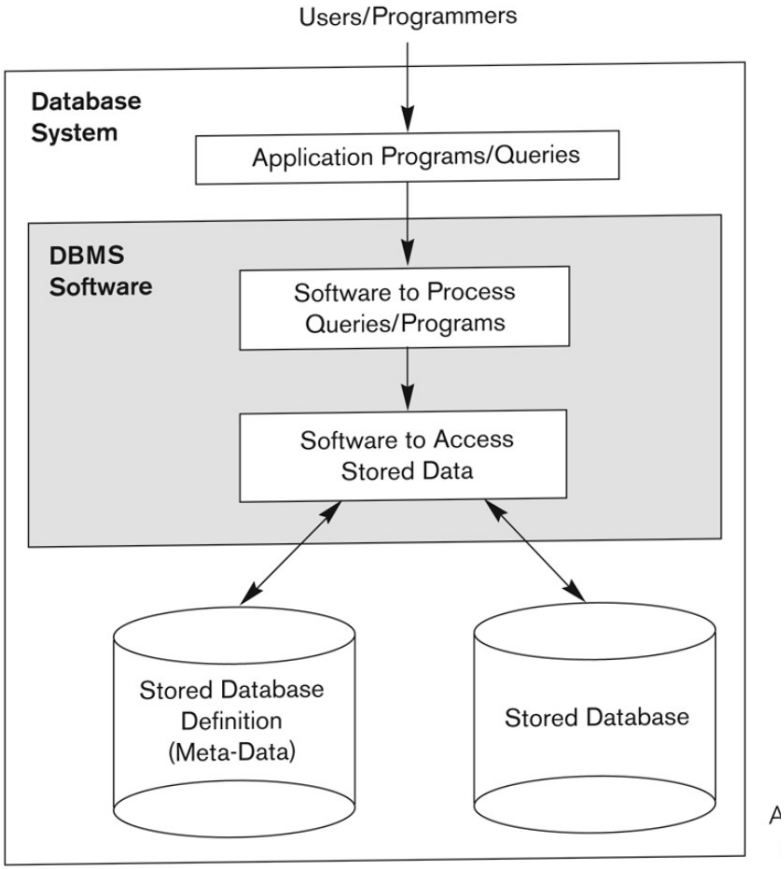
Example - Database

Database(DB) is of any size and complexity

- List of names and addresses
- Computerized catlog of large library
- Facebook - MySQL (Relational) → core user data & Cassandra / HBase (NoSQL) → messages, feeds
- Amazon - Aurora / RDS (Relational) → orders, payments & DynamoDB (NoSQL) → shopping cart, sessions

The DB may be maintained manually or it may be computerized.

Simplified Database Environment



Database Management System

- **Data Base Management System** is a collection of programs that enables users to create and maintain the database.
- A general-purpose software system that facilitates the processes of **defining, constructing, manipulating and sharing databases** for various applications.
  - **Defining a database** - Involves specifying the data types, structures and constraints for the data to be stored in the database. The database definition is also stored by the DBMS in the form of a database catalog or dictionary; it is called **meta-data**.
  - **Constructing a database** - Process of storing the data itself on a storage medium that is controlled by DBMS.
  - **Manipulating a database** - Includes functions such as querying the DB for retrieval, updating the database to reflect changes in the miniworld and generating the reports, etc.,
  - **Sharing**- Allows multiple users and programs to acess the database simultaneously

Any application program can access the database by sending queries or requests for data to the DBMS.

DBMS -Functions

Other functions provided by DBMS:

- **Protection:** includes system protection against hardware or software malfunction (or crashes) and security protection against unauthorized or malicious access.
- **Maintain:**A typical large database may have a long life cycle, the DBMS must be able to maintain the database system by allowing the system to evolve as requirements change over time.

The database and DBMS software together a database system.

Steps in Database Construction

The db design is implemented, populated with actual data and maintained to reflect the state of the mini world

- Specification and Analysis phase
- Conceptual Design - The requirements are transformed into a conceptual design using tools.
- Logical Design - Designing the relations using DBMS software.
- Physical Design - where further specifications are provided for storing and accessing the database.

Phase	How it is achieved	Output
Specification	Collect requirements	Document
Conceptual	ER modeling	Conceptual Schema
Logical	Convert ER $\rightarrow$ Relation	Logical Schema
Physical	Storage design	Physical Schema

Database - Example

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410

Figure 1.2
A database that stores

Characteristics of database Approach

Main Characteristics of database approach versus the file-processing approach

- Self-describing nature of a database system
- Insulation between programs and data and data abstraction
- Support of multiple views of the data
- Sharing of data and multi-user transaction processing

Characteristics of database Approach

Self-describing nature of a database system:

- A DBMS stores not only the data but also metadata (data about the data) is stored seprately in a catalog.
- Metadata includes:
  - Structure of files: Definition of tables, columns, data types, relationships, etc.
  - Storage formats: Information on how data are stored physically.
  - Constraints: Rules such as primary keys, foreign keys, and data integrity rules.
- In NoSQL database systems, explicit meta-data may not be required. Data is stored as self-describing data in a structure
- Each data item includes:
  - Attribute (field) names
  - Corresponding data values
- This information is used by both users and DBMS software to know about the database structure and to ensure consistency and enforce constraints during operations.

Characteristics of database Approach

- A general purpose DBMS software is
  - Designed to support multiple applications.
  - It relies on metadata in the catalog rather than being hardcoded for a specific database structure.
- Traditional file processing constrained to work with only one specific database whose structure is defined in the application programs.
  - Eg: An application program written in C++ may have struct or class declaration. File processing software can access only specific databases.
- DBMS software can access diverse databases by extracting the database definition from the catalog and using these definitions.
- This self-describing nature makes a DBMS versatile and powerful for managing modern database systems.

Database Catalog - Example

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....	....	.....
....	....	.....
....	....	.....
Prerequisite_number	XXXXNNNN	PREREQUISITE

Characteristics of database Approach

Insulation between programs and data and data abstraction:

- In file processing the structure of the files is embedded in the application programs any changes to the structure of the file may require changing all the programs that access the file.
- In DBMS the structure of the data files stored in the catalog is separated from the access programs.
- **Program data independence**: Changing the structure of the catalog file if necessary without changing the access programs.
- In object-oriented and object-relational systems, we can define operations on data as part of database definitions.
- **Program - operation Independence**: Application programs can operate on data by invoking operations (methods or functions) regardless of how operations are implemented.

Characteristics of database Approach

Insulation between programs and data and data abstraction:

- **Data Abstraction:** A characteristic that allow Program data independence and Program - operation Independence .
- DBMS provides users with conceptual representation of data that does not include details of how the data is stored or how the the operations are implemented.
- Data model is a type of data abstraction used to represent the conceptual representation.
- The data model uses logical concepts such as objects, their properties, and their inter relationship. This data model hides storage and implementation details that are not of interest to users.
- In object-oriented and object-relational databases, the abstraction process includes not only the data structure but also the operations on the data.

Characteristics of database Approach

Support of multiple views of the data:

- Each user may have a different perspective or view of the database, based on the data of interest to that user.
- Views are virtual data derived from one or more relation and or not explicitly stored.
- Many users may not aware the data they refer is stored or derived.
- A multiuser DBMS whose users have variety of distinct applications must provide facilities for defining multiple views.

Database views- Example

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

(a)

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

(b)

Characteristics of database Approach

Sharing of data and multi-user transaction processing:

- Allow multiple users to access the database at the same time correctly and efficiently.
- Concurrency control within the DBMS guarantees that each transaction is correctly executed or aborted.
- Eg: Several reservation agents try to assign a seat on an airline flight, the DBMS should ensure that each seat can be accessed by only one agent at a time for assignment to a passenger. These types of application are generally called online transaction processing (OLTP) applications.
- Transaction is an executing program or process that includes one or more database accesses such as reading and updating of database records.
- Transaction becomes central to many db applications and should enforce **ACID** properties

Actors on the scene

The people who design,use and administer a database

1. Data base Administrators: Administering the primary and secondary resources (DB and DBMS software)
   - Responsible for:
     - Authorizing access to the database
     - Coordinating and monitoring its use
     - Acquiring software and hardware resources
     - Accountable for problems such as security breaches and poor system response time.
   - Database Designers: Communicate with potential users to understand requirements
     - Create design to meet the requirements.
     - Identifying and choosing appropriate structures to represent and store this data.
     - Develop views of the database that meet the data and processing requirements of groups.
     - Each view is analyzed and integrated with the views of other user groups.

Actors on the Scene: End Users

Access the database for querying, updating, and generating reports
There are several categories of end users:

- **Casual End Users**:
  1. Occasionally access the database, need different information each time.
  2. Use a sophisticated database query interface to specify their requests.
  3. They are middle or high-level managers or occasional browsers.

Actors on the Scene: End Users

- **Navie or Parametric End Users**:
  1. Constantly querying and updating the database
  2. Uses standard types of queries and updates called **canned transactions**
  3. Learn very little about the facilities of DBMS
  4. Wider knowledge of user interfaces.
- **Sophisticated end users**
  1. Include engineers, scientists, business analysts, and others
  2. They are familiarized with the facilities of the DBMS.
  3. They are able to implement their own applications to meet their complex requirements.
- **Standalone users**:
  1. Maintain personal databases by using ready-made program packages interfaces.
  2. Proficient in using specific software package
  3. Eg: Financial software package that stores a variety of personal financial data.

System Analysts and Application Programmers

- **System analysts:**
  1. Determine the requirements of end users
  2. Develop specifications for standard canned transactions
- **Application programmers:**
  1. Implement the specifications as programs
  2. Test, debug, document, and maintain these canned transactions.

Workers behind the scene

People associated with the design, development, and operation of the DBMS software and system environment.

- **DBMS system designers and implementers:** Design and implement the DBMS modules and interfaces as a software package.
- **Tool developers design and implement tools:**—the software packages that facilitate database modeling and design, database system design, and improved performance.
- **Operators and maintenance personnel** (system administration personnel) responsible for the actual running and maintenance of the hardware and software environment for the database system.

Summary

- Introduction to DBMS
- Simplified example of Database
- Characteristics of database approach
- Database users

Reference

Fundamentals of Database systems 7th Edition by Ramez Elmasri.